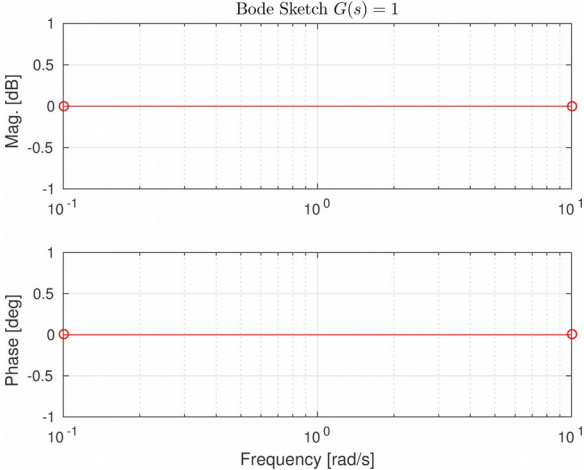
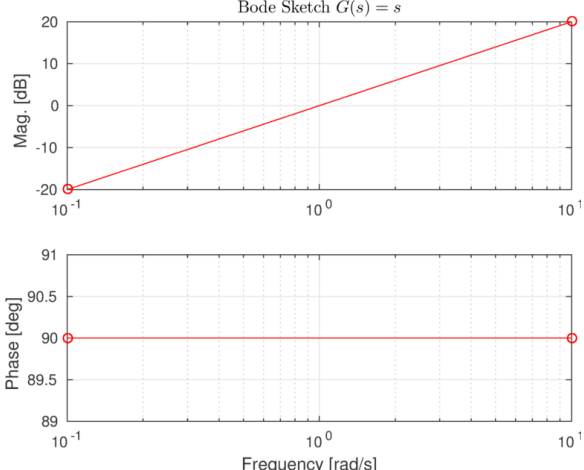
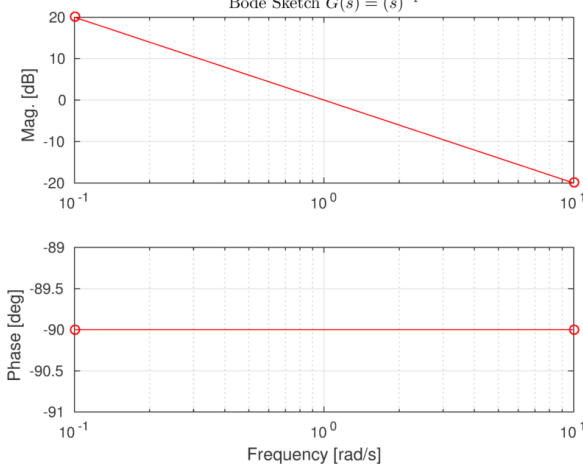


Graphical Bode Plot Building Blocks

G(s)	M = 20log(K) P=0 K Bode Sketch $G(s) = 1$ 
Derivative s	M = 20 dB/dec P = 90 deg Bode Sketch $G(s) = s$ 
Integral $\frac{1}{s}$	M = -20 dB/dec P = -90 deg Bode Sketch $G(s) = (s)^{-1}$ 

Real Zero $\tau s + 1$	M: - break at $\omega_c = 1/\tau$ - Const for $\omega < \omega_c$ - +20 dB/dec $\omega > \omega_c$ P: - 0 to +90 deg Bode Sketch $G(s) = (\tau s + 1)$, $\tau = 1$ [s]
Pole $\frac{1}{\tau s + 1}$	M: - break at $\omega_c = 1/\tau$ - Const for $\omega < \omega_c$ - (-20 dB/dec) $\omega > \omega_c$ P: - 0 to -90 deg Bode Sketch $G(s) = (\tau s + 1)^{-1}$, $\tau = 1$ [s]
Complex Zeros $(s/\omega_n)^2 + (2\zeta/\omega_n)s + 1$	M: - break at $\omega_c = \omega_n$ - Const for $\omega < \omega_c$ - (+40 dB/dec) $\omega > \omega_c$ P: - 0 to +180 deg Bode Sketch $G(s) = ((s/\omega_n)^2 + 2\zeta/\omega_n s + 1)$; $\omega_n = 1$, $\zeta = 0.50$
Complex Poles $\frac{1}{(s/\omega_n)^2 + (2\zeta/\omega_n)s + 1}$	M: - break at $\omega_c = \omega_n$ - Const for $\omega < \omega_c$ - (-40 dB/dec) $\omega > \omega_c$ P: - 0 to -180 deg Bode Sketch $G(s) = ((s/\omega_n)^2 + 2\zeta/\omega_n s + 1)^{-1}$; $\omega_n = 1$, $\zeta = 0.50$